Libtasn1

Abstract Syntax Notation One (ASN.1) library for the GNU system
part of the GnuTLS project
for version 1.3, 1 February 2008

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This manual is for Libtasn1 (version 1.3, 1 February 2008), which is a library for Abstract Syntax Notation One (ASN.1) and Distinguish Encoding Rules (DER) manipulation.

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1 Introduction

This document describes the Libtasn1 library developed for ASN.1 (Abstract Syntax Notation One) structures management and DER (Distinguished Encoding Rules) encoding functions.

The main features of this library are:

- On line ASN1 structure management that doesn't require any C code file generation.
- Off line ASN1 structure management with C code file generation containing an array.
- DER (Distinguish Encoding Rules) encoding.
- No limits for INTEGER and ENUMERATED values.
- It's Free Software. Anybody can use, modify, and redistribute the library under the terms of the GNU Lesser General Public License version 2.1 (see Section A.2 [GNU LGPL], page 25). The command line tools, self-tests and build infrastructure are licensed under the GNU General Public License version 3.0 (see Section A.3 [GNU GPL], page 33).
- It's thread-safe. No global variables are used and multiple library handles and session handles may be used in parallel.
- It's portable. It should work on all Unix like operating systems, including Windows. The library itself should be portable to any C89 system, not even POSIX is required.

2 ASN.1 structure handling

2.1 ASN.1 syntax

The parser is case sensitive. The comments begin with "– " and end at the end of lines. An example is in "pkix.asn" file. ASN.1 definitions must have this syntax:

```
definitions_name {<object definition>}

DEFINITIONS <EXPLICIT or IMPLICIT> TAGS ::=

BEGIN

<type and constants definitions>

END
```

The token "::=" must be separate from others elements, so this is a wrong declaration:

```
;; INCORRECT
Version ::=INTEGER
```

the correct form is:

```
Version ::= INTEGER
```

Here is the list of types that the parser can manage:

- INTEGER
- ENUMERATED
- BOOLEAN
- OBJECT IDENTIFIER
- NULL
- BIT STRING
- OCTET STRING
- UTCTime
- GeneralizedTime
- GeneralString
- SEQUENCE
- SEQUENCE OF
- SET
- SET OF
- CHOICE
- ANY
- ANY DEFINED BY

This version doesn't manage REAL type. It doesn't allow the "EXPORT" and "IMPORT" sections too.

The SIZE constraints are allowed, but no check is done on them.

2.2 Naming

Consider this definition:

```
Example { 1 2 3 4 }

DEFINITIONS EXPLICIT TAGS ::=

BEGIN

Group ::= SEQUENCE {
   id   OBJECT IDENTIFIER,
   value  Value
}

Value ::= SEQUENCE {
   value1  INTEGER,
   value2  BOOLEAN
}

END
```

To identify the type 'Group' you have to use the null terminated string "Example.Group". These strings are used in functions that are described below.

Others examples:

Field 'id' in 'Group' type : "Example.Group.id".

Field 'value1' in field 'value' in type 'Group': "Example.Group.value.value1".

Elements of structured types that don't have a name, receive the name "?1","?2", and so on.

The name "?LAST" indicates the last element of a `SET_OF` or `SEQUENCE_OF`.

2.3 Library Notes

The header file of this library is '`libtasn1.h`'.

The main type used in it is `ASN1_TYPE`, and it's used to store the ASN.1 definitions and structures (instances).

The constant ASN1_TYPE_EMPTY can be used for the variable initialization. For example:

```
ASN1_TYPE definitions=ASN1_TYPE_EMPTY;
```

Some functions require a parameter named errorDescription of char* type. The array must be already allocated and must have at least `MAX_ERROR_DESCRIPTION_SIZE` bytes (E.g, as in `char Description[MAX_ERROR_DESCRIPTION_SIZE];`).

`MAX_NAME_SIZE` indicates the maximum number of characters of a name inside a file with ASN1 definitions.

2.4 Future developments

- Add functions for a C code file generation containing equivalent data structures (not a single array like now).
- Type REAL.

3 Utilities

3.1 Invoking asn1Parser

'`asn1Parser`' reads one file with ASN1 definitions and generates a file with an array to use with libtasn1 functions.

```
Usage:  asn1Parser [options] file

Options:
 -h : shows the help message.
 -v : shows version information and exit.
 -c : checks the syntax only.
 -o file : output file.
 -n name : array name.
```

3.2 Invoking asn1Coding

'`asn1Coding`' generates a DER encoding from a file with ASN1 definitions and another one with assignments.

The file with assignments must have this syntax:

```
InstanceName  Asn1Definition

nameString  value

nameString  value
...
```

The output file is a binary file with the DER encoding.

```
Usage:  asn1Coding [options] file1 file2
 file1 : file with ASN1 definitions.
 file2 : file with assignments.
Options:
 -h : shows the help message.
 -v : shows version information and exit.
 -c : checks the syntax only.
 -o file : output file.
```

3.3 Invoking asn1Decoding

'`asn1Decoding`' generates an ASN1 structure from a file with ASN1 definitions and a binary file with a DER encoding.

```
Usage:  asn1Decoding [options] file1 file2 type
 file1 : file with ASN1 definitions.
 file2 : binary file with a DER encoding.
 type : ASN1 definition name.
Options:
 -h : shows the help message.
```

```
-v : shows version information and exit.
-c : checks the syntax only.
-o file : output file.
```

4 Function reference

4.1 ASN.1 schema functions

asn1_parser2tree

`asn1_retCode asn1_parser2tree` (*const char * `file_name`*, *ASN1_TYPE * `definitions`*, *char * `errorDescription`*) [Function]

file_name: specify the path and the name of file that contains ASN.1 declarations.

definitions: return the pointer to the structure created from "file_name" ASN.1 declarations.

errorDescription: return the error description or an empty string if success.

Creates the structures needed to manage the definitions included in *FILE_NAME file.

Returns: ASN1_SUCCESS: The file has a correct syntax and every identifier is known.

ASN1_ELEMENT_NOT_EMPTY: *POINTER not ASN1_TYPE_EMPTY.

ASN1_FILE_NOT_FOUND: An error occured while opening FILE_NAME.

ASN1_SYNTAX_ERROR: The syntax is not correct.

ASN1_IDENTIFIER_NOT_FOUND: In the file there is an identifier that is not defined.

ASN1_NAME_TOO_LONG: In the file there is an identifier whith more than MAX_NAME_SIZE characters.

asn1_parser2array

`int asn1_parser2array` (*const char * `inputFileName`*, *const char * `outputFileName`*, *const char * `vectorName`*, *char * `errorDescription`*) [Function]

inputFileName: specify the path and the name of file that contains ASN.1 declarations.

outputFileName: specify the path and the name of file that will contain the C vector definition.

vectorName: specify the name of the C vector.

errorDescription: return the error description or an empty string if success.

Creates a file containing a C vector to use to manage the definitions included in *INPUTFILENAME file. If *INPUTFILENAME is "/aa/bb/xx.yy" and OUTPUTFILENAME is NULL, the file created is "/aa/bb/xx_asn1_tab.c". If VECTORNAME is NULL the vector name will be "xx_asn1_tab".

Returns: ASN1_SUCCESS: The file has a correct syntax and every identifier is known.

ASN1_FILE_NOT_FOUND: An error occured while opening FILE_NAME.

ASN1_SYNTAX_ERROR: The syntax is not correct.

ASN1_IDENTIFIER_NOT_FOUND: In the file there is an identifier that is not defined.

ASN1_NAME_TOO_LONG: In the file there is an identifier whith more than MAX_NAME_SIZE characters.

4.2 ASN.1 field functions

asn1_array2tree

`asn1_retCode asn1_array2tree` (*const ASN1_ARRAY_TYPE * `array`, ASN1_TYPE * `definitions`, char * `errorDescription`*) [Function]

array: specify the array that contains ASN.1 declarations

definitions: return the pointer to the structure created by *ARRAY ASN.1 declarations

errorDescription: return the error description.

Creates the structures needed to manage the ASN.1 definitions. `array` is a vector created by `asn1_parser2array()`.

Returns: ASN1_SUCCESS: Structure created correctly.

ASN1_ELEMENT_NOT_EMPTY: *`definitions` not ASN1_TYPE_EMPTY.

ASN1_IDENTIFIER_NOT_FOUND: In the file there is an identifier that is not defined (see `errorDescription` for more information).

ASN1_ARRAY_ERROR: The array pointed by `array` is wrong.

asn1_delete_structure

`asn1_retCode asn1_delete_structure` (*ASN1_TYPE * `structure`*) [Function]

structure: pointer to the structure that you want to delete.

Deletes the structure *`structure`. At the end, *`structure` is set to ASN1_TYPE_EMPTY.

Returns: ASN1_SUCCESS: Everything OK.

ASN1_ELEMENT_NOT_FOUND: *`structure` was ASN1_TYPE_EMPTY.

asn1_delete_element

`asn1_retCode asn1_delete_element` (*ASN1_TYPE `structure`, const char * `element_name`*) [Function]

structure: pointer to the structure that contains the element you want to delete.

element_name: element's name you want to delete.

Deletes the element named *`element_name` inside *`structure`.

Returns: ASN1_SUCCESS: Everything OK.

ASN1_ELEMENT_NOT_FOUND: The name element was not found.

asn1_create_element

`asn1_retCode asn1_create_element` (*ASN1_TYPE `definitions`, const char * `source_name`, ASN1_TYPE * `element`*) [Function]

definitions: pointer to the structure returned by "parser_asn1" function

source_name: the name of the type of the new structure (must be inside p_structure).

element: pointer to the structure created.

Creates a structure of type `source_name`. Example using "pkix.asn":

rc = asn1_create_structure(cert_def, "PKIX1.Certificate", certptr);

Returns: ASN1_SUCCESS: Creation OK.

ASN1_ELEMENT_NOT_FOUND: SOURCE_NAME isn't known

asn1_print_structure

`void asn1_print_structure (`*FILE * out, ASN1_TYPE structure, const char * name, int mode*`)` [Function]

out: pointer to the output file (e.g. stdout).

structure: pointer to the structure that you want to visit.

name: an element of the structure

mode: specify how much of the structure to print, can be `ASN1_PRINT_NAME`, `ASN1_PRINT_NAME_TYPE`, `ASN1_PRINT_NAME_TYPE_VALUE`, or `ASN1_PRINT_ALL`.

Prints on the `out` file descriptor the structure's tree starting from the `name` element inside the structure `structure`.

asn1_number_of_elements

`asn1_retCode asn1_number_of_elements (`*ASN1_TYPE element, const char * name, int * num*`)` [Function]

element: pointer to the root of an ASN1 structure.

name: the name of a sub-structure of ROOT.

num: pointer to an integer where the result will be stored

Counts the number of elements of a sub-structure called NAME with names equal to "?1","?2", ...

Returns: ASN1_SUCCESS: Creation OK.

ASN1_ELEMENT_NOT_FOUND: NAME isn't known.

ASN1_GENERIC_ERROR: Pointer num equal to NULL.

asn1_find_structure_from_oid

`const char * asn1_find_structure_from_oid (`*ASN1_TYPE definitions, const char * oidValue*`)` [Function]

definitions: ASN1 definitions

oidValue: value of the OID to search (e.g. "1.2.3.4").

Search the structure that is defined just after an OID definition.

Returns: NULL when OIDVALUE not found, otherwise the pointer to a constant string that contains the element name defined just after the OID.

asn1_copy_node

`asn1_retCode asn1_copy_node (`*ASN1_TYPE dst, const char * dst_name, ASN1_TYPE src, const char * src_name*`)` [Function]

dst: Destination ASN1_TYPE node.

dst_name: Field name in destination node.

src: Source ASN1_TYPE node.

src_name: Field name in source node.

Create a deep copy of a ASN1_TYPE variable.

Return value: Return ASN1_SUCCESS on success.

asn1_write_value

`asn1_retCode asn1_write_value` (*ASN1_TYPE* `node_root`, *const char ** `name`, *const void ** `ivalue`, *int* `len`) [Function]

node_root: pointer to a structure

name: the name of the element inside the structure that you want to set.

ivalue: vector used to specify the value to set. If len is >0, VALUE must be a two's complement form integer. if len=0 *VALUE must be a null terminated string with an integer value.

len: number of bytes of *value to use to set the value: value[0]..value[len-1] or 0 if value is a null terminated string

Set the value of one element inside a structure.

If an element is OPTIONAL and you want to delete it, you must use the value=NULL and len=0. Using "pkix.asn":

result=asn1_write_value(cert, "tbsCertificate.issuerUniqueID", NULL, 0);

Description for each type: INTEGER: VALUE must contain a two's complement form integer.

value[0]=0xFF , len=1 -> integer=-1. value[0]=0xFF value[1]=0xFF , len=2 -> integer=-1. value[0]=0x01 , len=1 -> integer= 1. value[0]=0x00 value[1]=0x01 , len=2 -> integer= 1. value="123" , len=0 -> integer= 123.

ENUMERATED: As INTEGER (but only with not negative numbers).

BOOLEAN: VALUE must be the null terminated string "TRUE" or "FALSE" and LEN != 0.

value="TRUE" , len=1 -> boolean=TRUE. value="FALSE" , len=1 -> boolean=FALSE.

OBJECT IDENTIFIER: VALUE must be a null terminated string with each number separated by a dot (e.g. "1.2.3.543.1"). LEN != 0.

value="1 2 840 10040 4 3" , len=1 -> OID=dsa-with-sha.

UTCTime: VALUE must be a null terminated string in one of these formats: "YYMMDDhhmmssZ", "YYMMDDhhmmssZ", "YYMMDDhhmmss+hh'mm'", "YYMMDDhhmmss-hh'mm'", "YYMMDDhhmm+hh'mm'", or "YYMMDDhhmm-hh'mm'". LEN != 0.

value="9801011200Z" , len=1 -> time=Jannuary 1st, 1998 at 12h 00m Greenwich Mean Time

GeneralizedTime: VALUE must be in one of this format: "YYYYMMDDhhmmss.sZ", "YYYYMMDDhhmmss.sZ", "YYYYMMDDhhmmss.s+hh'mm'", "YYYYMMDDhhmmss.s-hh'mm'", "YYYYMMDDhhmm+hh'mm'", or

"YYYYMMDDhhmm-hh'mm'" where ss.s indicates the seconds with any precision like "10.1" or "01.02". LEN != 0

value="2001010112001.12-0700" , len=1 -> time=Jannuary 1st, 2001 at 12h 00m 01.12s Pacific Daylight Time

OCTET STRING: VALUE contains the octet string and LEN is the number of octets.

value="$\backslash$x01$\backslash$x02$\backslash$x03" , len=3 -> three bytes octet string

GeneralString: VALUE contains the generalstring and LEN is the number of octets.

value="$\backslash$x01$\backslash$x02$\backslash$x03" , len=3 -> three bytes generalstring

BIT STRING: VALUE contains the bit string organized by bytes and LEN is the number of bits.

value="$\backslash$xCF" , len=6 -> bit string="110011" (six bits)

CHOICE: if NAME indicates a choice type, VALUE must specify one of the alternatives with a null terminated string. LEN != 0. Using "pkix.asn"\:

result=asn1_write_value(cert, "certificate1.tbsCertificate.subject", "rdnSequence", 1);

ANY: VALUE indicates the der encoding of a structure. LEN != 0.

SEQUENCE OF: VALUE must be the null terminated string "NEW" and LEN != 0. With this instruction another element is appended in the sequence. The name of this element will be "?1" if it's the first one, "?2" for the second and so on.

Using "pkix.asn"\:

result=asn1_write_value(cert, "certificate1.tbsCertificate.subject.rdnSequence", "NEW", 1);

SET OF: the same as SEQUENCE OF. Using "pkix.asn":

result=asn1_write_value(cert, "tbsCertificate.subject.rdnSequence.?LAST", "NEW", 1);

Returns: ASN1_SUCCESS: Set value OK.

ASN1_ELEMENT_NOT_FOUND: NAME is not a valid element.

ASN1_VALUE_NOT_VALID: VALUE has a wrong format.

asn1_read_value

`asn1_retCode asn1_read_value` (*ASN1_TYPE* `root`, *const char ** `name`, *void ** `ivalue`, *int ** `len`) [Function]

root: pointer to a structure.

name: the name of the element inside a structure that you want to read.

ivalue: vector that will contain the element's content, must be a pointer to memory cells already allocated.

len: number of bytes of *value: value[0]..value[len-1]. Initialy holds the sizeof value.

Returns the value of one element inside a structure.

If an element is OPTIONAL and the function "read_value" returns ASN1_ELEMENT_NOT_FOUND, it means that this element wasn't present in the der encoding that created the structure. The first element of a SEQUENCE_OF or SET_OF is named "?1". The second one "?2" and so on.

INTEGER: VALUE will contain a two's complement form integer.

integer=-1 -> value[0]=0xFF , len=1. integer=1 -> value[0]=0x01 , len=1.

ENUMERATED: As INTEGER (but only with not negative numbers).

BOOLEAN: VALUE will be the null terminated string "TRUE" or "FALSE" and LEN=5 or LEN=6.

OBJECT IDENTIFIER: VALUE will be a null terminated string with each number separated by a dot (i.e. "1.2.3.543.1").

LEN = strlen(VALUE)+1

UTCTime: VALUE will be a null terminated string in one of these formats: "YYMMDDhhmmss+hh'mm'" or "YYMMDDhhmmss-hh'mm'". LEN=strlen(VALUE)+1.

GeneralizedTime: VALUE will be a null terminated string in the same format used to set the value.

OCTET STRING: VALUE will contain the octet string and LEN will be the number of octets.

GeneralString: VALUE will contain the generalstring and LEN will be the number of octets.

BIT STRING: VALUE will contain the bit string organized by bytes and LEN will be the number of bits.

CHOICE: If NAME indicates a choice type, VALUE will specify the alternative selected.

ANY: If NAME indicates an any type, VALUE will indicate the DER encoding of the structure actually used.

Returns: ASN1_SUCCESS: Set value OK.

ASN1_ELEMENT_NOT_FOUND: NAME is not a valid element.

ASN1_VALUE_NOT_FOUND: There isn't any value for the element selected.

ASN1_MEM_ERROR: The value vector isn't big enough to store the result. In this case LEN will contain the number of bytes needed.

asn1_read_tag

`asn1_retCode asn1_read_tag (`*node_asn * **root***, *const char * **name***, *int * **tagValue***, *int * **classValue***)` [Function]

root: pointer to a structure

name: the name of the element inside a structure.

tagValue: variable that will contain the TAG value.

classValue: variable that will specify the TAG type.

Returns the TAG and the CLASS of one element inside a structure.

CLASS can have one of these constants: ASN1_CLASS_APPLICATION, ASN1_CLASS_UNIVERSAL, ASN1_CLASS_PRIVATE or ASN1_CLASS_CONTEXT_SPECIFIC.

Returns: ASN1_SUCCESS: Set value OK.
ASN1_ELEMENT_NOT_FOUND: NAME is not a valid element.

4.3 DER functions

asn1_length_der

`void asn1_length_der (`*unsigned long int* `len`, *unsigned char ** `ans`, *int ** `ans_len``)` [Function]

len: value to convert.

ans: string returned.

ans_len: number of meaningful bytes of ANS (ans[0]..ans[ans_len-1]).

Creates the DER coding for the LEN parameter (only the length). The `ans` buffer is pre-allocated and must have room for the output.

asn1_octet_der

`void asn1_octet_der (`*const unsigned char ** `str`, *int* `str_len`, *unsigned char ** `der`, *int ** `der_len``)` [Function]

str: OCTET string.

str_len: STR length (str[0]..str[str_len-1]).

der: string returned.

der_len: number of meaningful bytes of DER (der[0]..der[ans_len-1]).

Creates the DER coding for an OCTET type (length included).

asn1_bit_der

`void asn1_bit_der (`*const unsigned char ** `str`, *int* `bit_len`, *unsigned char ** `der`, *int ** `der_len``)` [Function]

str: BIT string.

bit_len: number of meaningful bits in STR.

der: string returned.

der_len: number of meaningful bytes of DER (der[0]..der[ans_len-1]).

Creates the DER coding for a BIT STRING type (length and pad included).

asn1_der_coding

`asn1_retCode asn1_der_coding (`*ASN1_TYPE* `element`, *const char ** `name`, *void ** `ider`, *int ** `len`, *char ** `ErrorDescription``)` [Function]

element: pointer to an ASN1 element

name: the name of the structure you want to encode (it must be inside *POINTER).

ider: vector that will contain the DER encoding. DER must be a pointer to memory cells already allocated.

len: number of bytes of *`ider`: `ider`[0]..`ider`[len-1], Initialy holds the sizeof of der vector.

Creates the DER encoding for the NAME structure (inside *POINTER structure).

Returns: ASN1_SUCCESS: DER encoding OK.

ASN1_ELEMENT_NOT_FOUND: NAME is not a valid element.

ASN1_VALUE_NOT_FOUND: There is an element without a value.

ASN1_MEM_ERROR: `ider` vector isn't big enough. Also in this case LEN will contain the length needed.

asn1_get_length_der

`signed long asn1_get_length_der` (*const unsigned char * `der`, int `der_len`, int * `len`*) [Function]

der: DER data to decode.

der_len: Length of DER data to decode.

len: Output variable containing the length of the DER length field.

Extract a length field from DER data.

Return value: Return the decoded length value, or -1 on indefinite length, or -2 when the value was too big.

asn1_get_tag_der

`int asn1_get_tag_der` (*const unsigned char * `der`, int `der_len`, unsigned char * `cls`, int * `len`, unsigned long * `tag`*) [Function]

der: DER data to decode.

der_len: Length of DER data to decode.

cls: Output variable containing decoded class.

len: Output variable containing the length of the DER TAG data.

tag: Output variable containing the decoded tag.

Decode the class and TAG from DER code.

Return value: Returns ASN1_SUCCESS on success, or an error.

asn1_get_octet_der

`int asn1_get_octet_der` (*const unsigned char * `der`, int `der_len`, int * `ret_len`, unsigned char * `str`, int `str_size`, int * `str_len`*) [Function]

der: DER data to decode containing the OCTET SEQUENCE.

der_len: Length of DER data to decode.

ret_len: Output variable containing the length of the DER data.

str: Pre-allocated output buffer to put decoded OCTET SEQUENCE in.

str_size: Length of pre-allocated output buffer.

str_len: Output variable containing the length of the OCTET SEQUENCE.

Extract an OCTET SEQUENCE from DER data.

Return value: Returns ASN1_SUCCESS on success, or an error.

asn1_get_bit_der

`int asn1_get_bit_der` (*const unsigned char * der, int der_len, int * ret_len, unsigned char * str, int str_size, int * bit_len*) [Function]

der: DER data to decode containing the BIT SEQUENCE.

der_len: Length of DER data to decode.

ret_len: Output variable containing the length of the DER data.

str: Pre-allocated output buffer to put decoded BIT SEQUENCE in.

str_size: Length of pre-allocated output buffer.

bit_len: Output variable containing the size of the BIT SEQUENCE.

Extract a BIT SEQUENCE from DER data.

Return value: Return ASN1_SUCCESS on success, or an error.

asn1_der_decoding

`asn1_retCode asn1_der_decoding` (*ASN1_TYPE * element, const void * ider, int len, char * errorDescription*) [Function]

element: pointer to an ASN1 structure.

ider: vector that contains the DER encoding.

len: number of bytes of *`ider`: `ider`[0]..`ider`[len-1].

errorDescription: null-terminated string contains details when an error occurred.

Fill the structure *ELEMENT with values of a DER encoding string. The sructure must just be created with function 'create_stucture'. If an error occurs during the decoding procedure, the *ELEMENT is deleted and set equal to `ASN1_TYPE_EMPTY`.

Returns: ASN1_SUCCESS: DER encoding OK.

ASN1_ELEMENT_NOT_FOUND: ELEMENT is ASN1_TYPE_EMPTY.

ASN1_TAG_ERROR,ASN1_DER_ERROR: The der encoding doesn't match the structure NAME. *ELEMENT deleted.

asn1_der_decoding_element

`asn1_retCode asn1_der_decoding_element` (*ASN1_TYPE * structure, const char * elementName, const void * ider, int len, char * errorDescription*) [Function]

structure: pointer to an ASN1 structure

elementName: name of the element to fill

ider: vector that contains the DER encoding of the whole structure.

len: number of bytes of *der: der[0]..der[len-1]

errorDescription: null-terminated string contains details when an error occurred.

Fill the element named ELEMENTNAME with values of a DER encoding string. The sructure must just be created with function 'create_stucture'. The DER vector must contain the encoding string of the whole STRUCTURE. If an error occurs during the decoding procedure, the *STRUCTURE is deleted and set equal to `ASN1_TYPE_EMPTY`.

Returns: ASN1_SUCCESS: DER encoding OK.

ASN1_ELEMENT_NOT_FOUND: ELEMENT is ASN1_TYPE_EMPTY or elementName == NULL.

ASN1_TAG_ERROR,ASN1_DER_ERROR: The der encoding doesn't match the structure STRUCTURE. *ELEMENT deleted.

asn1_der_decoding_startEnd

`asn1_retCode asn1_der_decoding_startEnd` (*ASN1_TYPE* `element`, *const void ** `ider`, *int* `len`, *const char ** `name_element`, *int ** `start`, *int ** `end`) [Function]

element: pointer to an ASN1 element

ider: vector that contains the DER encoding.

len: number of bytes of *`ider`: `ider`[0]..`ider`[len-1]

name_element: an element of NAME structure.

start: the position of the first byte of NAME_ELEMENT decoding (`ider`[*start])

end: the position of the last byte of NAME_ELEMENT decoding (`ider`[*end])

Find the start and end point of an element in a DER encoding string. I mean that if you have a der encoding and you have already used the function "asn1_der_decoding" to fill a structure, it may happen that you want to find the piece of string concerning an element of the structure.

Example: the sequence "tbsCertificate" inside an X509 certificate.

Returns: ASN1_SUCCESS: DER encoding OK.

ASN1_ELEMENT_NOT_FOUND: ELEMENT is ASN1_TYPE EMPTY or NAME_ELEMENT is not a valid element.

ASN1_TAG_ERROR,ASN1_DER_ERROR: the der encoding doesn't match the structure ELEMENT.

asn1_expand_any_defined_by

`asn1_retCode asn1_expand_any_defined_by` (*ASN1_TYPE* `definitions`, *ASN1_TYPE ** `element`) [Function]

definitions: ASN1 definitions

element: pointer to an ASN1 structure

Expands every "ANY DEFINED BY" element of a structure created from a DER decoding process (asn1_der_decoding function). The element ANY must be defined by an OBJECT IDENTIFIER. The type used to expand the element ANY is the first one following the definition of the actual value of the OBJECT IDENTIFIER.

Returns: ASN1_SUCCESS: Substitution OK.

ASN1_ERROR_TYPE_ANY: Some "ANY DEFINED BY" element couldn't be expanded due to a problem in OBJECT_ID -> TYPE association.

other errors: Result of der decoding process.

asn1_expand_octet_string

`asn1_retCode asn1_expand_octet_string` (*ASN1_TYPE* `definitions`, *ASN1_TYPE* * `element`, *const char* * `octetName`, *const char* * `objectName`) [Function]

definitions: ASN1 definitions

element: pointer to an ASN1 structure

octetName: name of the OCTECT STRING field to expand.

objectName: name of the OBJECT IDENTIFIER field to use to define the type for expansion.

Expands an "OCTET STRING" element of a structure created from a DER decoding process (asn1_der_decoding function). The type used for expansion is the first one following the definition of the actual value of the OBJECT IDENTIFIER indicated by OBJECTNAME.

Returns: ASN1_SUCCESS: Substitution OK.

ASN1_ELEMENT_NOT_FOUND: OBJECTNAME or OCTETNAME are not correct.

ASN1_VALUE_NOT_VALID: Wasn't possible to find the type to use for expansion.

other errors: result of der decoding process.

4.4 Error handling functions

libtasn1_perror

`void libtasn1_perror` (*asn1_retCode* `error`) [Function]

error: is an error returned by a libtasn1 function.

This function is like `perror()`. The only difference is that it accepts an error returned by a libtasn1 function.

libtasn1_strerror

`const char * libtasn1_strerror` (*asn1_retCode* `error`) [Function]

error: is an error returned by a libtasn1 function.

This function is similar to `strerror()`. The only difference is that it accepts an error (number) returned by a libtasn1 function.

Returns: Pointer to static zero-terminated string describing error code.

4.5 Auxilliary functions

asn1_find_node

`ASN1_TYPE asn1_find_node` (*ASN1_TYPE* `pointer`, *const char* * `name`) [Function]

pointer: NODE_ASN element pointer.

name: null terminated string with the element's name to find.

Searches for an element called NAME starting from POINTER. The name is composed by differents identifiers separated by dots. When *POINTER has a name, the first identifier must be the name of *POINTER, otherwise it must be the name of one child of *POINTER.

Return value: the searching result. NULL if not found.

asn1_check_version

`const char * asn1_check_version` (*const char * `req_version`*) [Function]

req_version: Required version number, or NULL.

Check that the the version of the library is at minimum the requested one and return the version string; return `NULL` if the condition is not satisfied. If a `NULL` is passed to this function, no check is done, but the version string is simply returned.

See `LIBTASN1_VERSION` for a suitable `req_version` string.

Return value: Version string of run-time library, or `NULL` if the run-time library does not meet the required version number.

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   When a "work that uses the Library" uses material from a header file that is part of the Library, the object code for the work may be a derivative work of the Library even though the source code is not. Whether this is true is especially significant if the work can be linked without the Library, or if the work is itself a library. The threshold for this to be true is not precisely defined by law.

   If such an object file uses only numerical parameters, data structure layouts and accessors, and small macros and small inline functions (ten lines or less in length), then the use of the object file is unrestricted, regardless of whether it is legally a derivative work. (Executables containing this object code plus portions of the Library will still fall under Section 6.)

   Otherwise, if the work is a derivative of the Library, you may distribute the object code for the work under the terms of Section 6. Any executables containing that work also fall under Section 6, whether or not they are linked directly with the Library itself.

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Version 3, 29 June 2007

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